BARQ

Report 2

Status and Progress

Aryaman Gupta · Krish Agarwal

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Abstract

BARQ is a twelve-degree-of-freedom quadruped robot. This report covers the development period since Report 1, during which the complete control, perception, autonomy and embedded-firmware stack was brought to maturity in high-fidelity physics simulation and validated end-to-end against an emulated flight controller. Every milestone planned for the pre-hardware phase is complete: the robot walks under velocity command, maps and navigates an unknown obstacle course autonomously, estimates its own motion without external reference, and exercises the exact embedded protocol it will use on metal — all with zero physical hardware attached. The simulator has been calibrated to the measured mass and the published actuator specification, making it a faithful digital twin rather than an idealisation. In parallel, a complete, self-contained execution plan from the current state to a fully functioning physical robot has been authored, substantially de-risking the build. Components are now arriving for physical assembly.

1 Executive summary

The programme strategy is to front-load engineering into simulation so that scarce and costly hardware iterations are minimised. That strategy has paid off in this period. The full software pipeline — velocity command, gait generation, exact inverse kinematics, motor control, and physics — runs in the Gazebo simulator with the robot translating forward, level and straight, and standing-height predicted by our model to within 0.2 mm of the simulated physics. On top of this foundation we have demonstrated autonomous navigation through a populated 16 m obstacle course, an onboard motion estimator that needs no external tracking, and a firmware-and-interface layer proven against an emulated controller to the point where commissioning the real electronics reduces to a single configuration change.

Crucially, the simulation is no longer idealised: it now carries the team's measured link masses and the manufacturer's actuator torque and speed limits, and the actuator response has been matched to the servo specification. Results obtained in simulation are therefore quantitatively meaningful — including the actuator load analysis presented in Section 2.7, which confirms the chosen servos carry the walking gait with a comfortable continuous safety margin.

2 Status and progress

2.1 Simulation and control foundation

The control stack was brought up in stages, each gated by a measurable fidelity check: kinematic visualisation, simulated motor control, analytical inverse kinematics, trot-gait generation, and finally full rigid-body physics with gravity and ground contact. A key correction in this period replaced an idealised leg model with one derived exactly from the robot's mechanical definition; this was identified through an adversarial multi-agent design review and eliminated a 3.4 cm foot-placement error that had reversed the walking direction. Following the fix, the model predicts the robot's settled standing height to within 0.2 mm of the physics — a fifty-fold improvement and our headline indicator that the simulation faithfully represents the machine.

2.2 Physical-fidelity calibration

The simulated robot was updated with the team's measured component masses (2.448 kg total, including the battery) and the published Waveshare ST3215 actuator limits: 2.94 N·m peak torque (30 kg·cm) and 4.71 rad/s no-load speed. The servo's internal control stiffness was matched to specification and the resulting joint-tracking error verified at 17.8 milliradians RMS during a walk. These steps convert the simulator from a qualitative animation into a quantitative digital twin whose force and torque outputs can be trusted (Section 2.7).

2.3 Locomotion performance

Under velocity command the robot walks forward, straight and level. It currently realises approximately 60% of the commanded forward speed in open-loop walking; the cause of the shortfall was diagnosed rigorously (a swing-foot ground-clearance limit, isolated by a controlled friction sweep) and is fully recoverable with the closed-loop feedback and learned-policy work planned for later phases. A deliberate stance trim shifts load toward the front feet for fore-aft stability. The gait was also shown to compose linear and turning commands correctly, tracing a clean closed circle of the expected radius.

2.4 Perception and autonomous navigation

A simulated 2-D laser scanner (matched to the specification of the intended sensor) feeds an industry-standard mapping and navigation stack. The robot built a map of an unknown environment and then completed a 16 metre autonomous mission through a course comprising a one-metre doorway, a three-pillar slalom and a scattered-obstacle field, finishing 0.157 m from the commanded waypoint and recovering from mid-course stalls without human intervention. Navigation speed is regulated automatically by proximity to obstacles and path curvature.

2.5 State estimation

An onboard estimator fuses leg kinematics with inertial measurement to estimate the robot's motion using no external reference — a prerequisite for real-world autonomy. Measured drift is 4–5% of distance travelled, within the normal band for kinematic legged odometry, and the mapping stack was shown to run successfully on this onboard estimate rather than on simulator ground truth.

2.6 Embedded firmware and hardware interface

The communication protocol between the onboard computer and the real-time microcontroller was implemented and cross-verified in both languages against shared reference test vectors. The robot's motor-control interface was then validated against a software emulator that runs the exact microcontroller firmware logic: the entire stack — from gait through to serial telemetry — passes a nine-point integration test and walks with no hardware present. Commissioning the physical electronics is consequently reduced to changing a single device address; the firmware itself is written and compiles for the target microcontroller.

2.7 Actuator load analysis

Using the calibrated digital twin, we measured the torque borne by each of the twelve servos throughout a normal walking cycle (Figure 1). The continuous (RMS) load peaks at 1.31 N·m — 45% of the actuator's rated capacity, a continuous safety factor of 2.2×. Brief torque transients at foot-strike approach the rated limit, concentrated on the rear legs; these are partly an artefact of perfectly-rigid simulated contact and will be re-measured on the bench, but they identify where to watch actuator current first on hardware. The analysis confirms the selected servos are appropriately sized for the walking gait, with margin, and provides the baseline against which future control improvements will be judged.

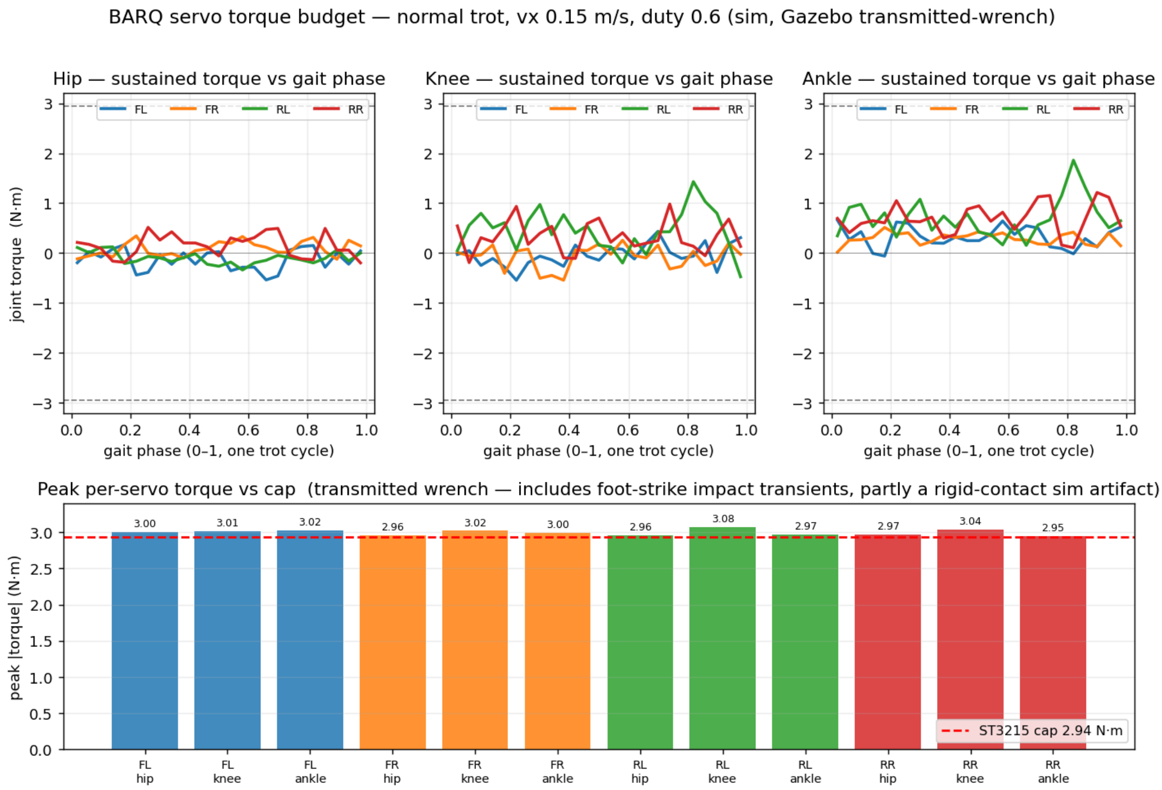

Figure 1. Per-servo torque during a normal trot (vx 0.15 m/s). Top: sustained torque vs. gait phase by joint class. Bottom: peak torque per servo against the 2.94 N·m actuator cap (dashed). Source: simulated joint transmitted-wrench, ground reaction included.

3 Quantitative results

The table below collects the period's primary measured indicators. All figures are reproducible from the project's versioned tooling and data logs.

Indicator	Result	Significance
Model fidelity (settle-height error)	0.2 mm	Simulation faithfully matches the modelled machine
Joint tracking error (walking)	17.8 mrad RMS	Actuator response calibrated to specification
Autonomous mission	16 m course completed	Final error 0.157 m; self-recovering
Onboard odometry drift	4–5% of distance	No external reference required
Hardware-interface integration	9 / 9 checks	Full stack runs without physical hardware
Protocol codec tests	6 / 6 + golden vectors	Computer–microcontroller link verified
Control / kinematics test suite	30 tests passing	Regression-protected

Continuous actuator load	45% of rated torque	2.2× continuous safety margin

4 Engineering process and risk management

Beyond technical results, the period produced durable process assets that reduce programme risk. Every engineering decision, change and experiment is recorded in a structured, version-controlled documentation system, with quantitative before/after metrics retained for eventual publication. The robot's power architecture has been specified and recorded (a single battery feeding a regulated motor supply, with a defined low-voltage protection ladder).

Most significantly, a complete execution plan — spanning calibration, mechanical assembly, firmware commissioning, first walking, perception, autonomy and machine learning, with measurable acceptance criteria and explicit contingency procedures at every step — has been authored as a self-contained reference. This plan is designed to be executed by a competent engineer without reliance on any single team member or tool, directly addressing key-person and continuity risk, and includes a fully specified fallback path to a deployable robot should the most advanced (machine-learning) approach prove unnecessary or unavailable.

5 Current status and next steps

Phase	Status
Simulation, control and inverse kinematics	Complete
Physics-fidelity calibration to measured hardware	Complete
Perception, SLAM and autonomous navigation	Complete (simulation)
Onboard state estimation	Complete (simulation)
Embedded protocol and hardware interface	Complete; validated on emulator
Physical assembly and commissioning	Ready; awaiting components
On-robot walking and autonomy	Planned; fully specified
Learned locomotion policy	Planned; three compute paths + fallback

With components now arriving, the immediate next steps are per-servo bench calibration, mechanical assembly, and firmware commissioning — each with pre-defined pass/fail gates and contingency procedures. Because the entire software stack is already proven against an emulated controller, the transition to physical hardware is expected to be incremental rather than exploratory. We will report measured on-robot performance against the simulation baselines established in this period in the next update.

— Aryaman Gupta & Krish Agarwal